

1.3.1. Introduction.

The theory of numbers, an important branch of mathematics, is primarily concerned with the properties of the natural numbers, 1, 2, 3, 4,, also called the positive integers. However, the theory is not strictly confined to just the natural numbers or even to the set of all integers: $0, \pm 1, \pm 2, \dots$. In fact, some theorems of number theory can easily be proved by applying the properties of real or complex numbers. Some important applications of number theory are in computer arithmetic that includes transmission, coding, manipulation of numerical data and also in cryptology—the study of secret messages. The aim of the present chapter is to introduce the divisibility theory and properties, greatest common divisors (GCD) and some properties of GCD with applications and congruence relations with a few applications.

1.3.2. Well ordering principle

Every non-empty subset of natural numbers has a smallest element. Thus, if S be a non-empty subset of natural numbers, then there exists $p \in S$ such that $p \leq q$ for all $q \in S$.

A set containing just one element has a smallest member, namely the element itself. Hence, the well-ordering principle is true for set of size 1.

Now let us assume that the principle is true for sets of size n , i.e., any set of n natural numbers has a smallest number

Let us now consider a set S of $(n+1)$ numbers from which one element 'p' is removed. The remaining n numbers have a smallest element say q (by the induction hypothesis). The smaller of p and q is the smallest element of S .

Hence, by the principle of **mathematical induction**, it follows that any non-empty finite set of natural numbers has a smallest element.

✓ 1.3.3. Divisibility theory

Definition. An integer b is divisible by an integer $a (\neq 0)$, if there is an integer x such that $b = ax$, and we write $a \mid b$.

In case b is not divisible by a , we write $a \nmid b$

Note. (i) when a divides b , a is called a divisor or factor at b and b is called a multiple of a

(ii) If a divides b then $-a$ also divides b because $b = ax \Rightarrow b = (-a)(-x)$

i.e., $a \mid b \Rightarrow -a \mid b$

Illustrations.

(i) 38 is divisible by 19 since $38 = 19 \times 2$

(ii) 11 is a divisor of 143 since $143 = 11 \times 13$

(iii) 0 is divisible by every integer because $0 = x \times 0$ for every value x .

1.3.4. Properties of divisibility

Some basic properties of divisibility of integers are given in the following theorem:

✓ **Theorem-1** Let $a, b, c \in \mathbb{Z}$, the set of integers. Then

- ✓ (i) $a \mid a \forall a \in \mathbb{Z}$ (Reflexive property)
- ✓ (ii) If $a \mid b$, then $a \mid bn$, for any integer n
- ✓ (iii) If $a \mid b$ and $b \mid c$, then $a \mid c$ (Transitivity property)
- ✓ (iv) If $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for any integers x and y (Linear property)
- ✓ (v) If $a \mid b$ then $xa \mid xb \forall x \in \mathbb{Z}$ (Multiplicative property)
- ✓ (vi) If $xa \mid xb$ then $a \mid b$ when $x \neq 0$ (cancellation law)
- ✓ (vii) If $a \mid b$, $b \neq 0$ then $|a| \leq |b|$ (comparison property)
- ✓ (viii) If $a \mid b$ and $b \mid a$, then $|a| = |b|$, i.e. $a = \pm b$ (comparison property)

Proof.(i) Since $a = 1 \cdot a$, $1 \in \mathbb{Z}$ This means a divides a

$$\therefore a|a \quad \forall a \in \mathbb{Z}$$

(ii) Since $a|b$, we have $b = ax$ for some integer x

$$\therefore bn = anx = (nx)a \text{ where } n, x \text{ are integers}$$

This means that a divides bn

$$\therefore a|bn$$

(iii) Since $a|b$ and $b|c$, we have

$$b = pa \text{ and } c = qb \text{ where } p, q \text{ are integers}$$

$$\therefore c = qb = q(pa) = (pq)a$$

This means that a divides c

$$\therefore a|c$$

(iv) Since $a|b$ and $a|c$, so we have

$$b = pa \text{ and } c = qa \text{ where } p, q \text{ are integers}$$

Now $bx + cy$

$$= (pa)x + (qa)y$$

$$= (px + qy)a$$

This means that a divides $bx + cy$, as p, q, x and y are integers.

$$\therefore a|bx + cy$$

(v) Since $a|b$, we have $b = ay$ for some integer y

$$\therefore bx = ayx = (ax)y$$

$$\therefore xa|xb$$

(vi) Since $xa|xb$, so we have

$$xb = (xa)y \text{ for some } y \in \mathbb{Z}$$

$$\therefore b = ay \quad (\because a \neq 0)$$

$$\therefore a|b$$

(vii) Since $a|b$, we have

$$b = pa \text{ for some integers } p$$

$$\therefore |b| = |pa| = |p||a|$$

$$\therefore |a| = \frac{|b|}{|p|} \leq |b| \quad [\because |p| \geq 1 \text{ and } p \neq 0]$$

$$\therefore |a| \leq |b|$$

Note: When $a > 0$, $b > 0$,

$$a|b \Rightarrow a \leq b$$

(viii) $a|b$ and $b|a$ implies

$$b = xa, \quad a = by \text{ for some } x, y \in \mathbb{Z}$$

$$\therefore b = xa = x.by$$

$$\Rightarrow xy = 1.$$

 $\therefore$ Either $x = 1, y = 1$ or $x = -1, y = -1$

$$\therefore b = a, \quad b = -a$$

$$\text{or, } a = b, \quad a = -b$$

Thus $a = \pm b$ i.e. $|a| = |b|$.**Corollary.** If $a|b$ and $a|c$, then

$$a|(b+c) \text{ and } a|(b-c)$$

Proof follows from (iv)

Prime numbers**Definition.** A positive integer $p > 1$ is called **prime** if the only positive factors of p are 1 and p . If a positive integer $n > 1$ is not prime, then n is called **composite****Note.** (1) The positive integer 1 is neither prime nor composite

(2) The only even prime integer is 2; rest all other primes are odd integers

(3) If n is a composite, then there exists positive integers a and b such that $n = ab$ where $1 < a, b < n$

Illustrations. (i) The integers 2, 3, 5, 7 etc are prime where as 4, 6, 9, 10 etc are composite since $4 = 2 \times 2$, $6 = 3 \times 2$, $9 = 3 \times 3$ and $10 = 5 \times 2$

1.3.5. Fundamental theorem of arithmetic

Every integer n greater than 1 can be expressed uniquely as a product of primes

Proof. If the integer $n > 1$ is a prime, then the integer itself stands as a product with a single factor. Otherwise n can be factored into, say, $n_1 n_2$, where $1 < n_1 < n$ and $1 < n_2 < n$. If n_1 is a prime, let it stand; otherwise it will be factor into, say, $n_3 n_4$ where $1 < n_3 < n_1$ and $1 < n_4 < n_1$; similarly for n_2 . This process of writing each composite number that arises as a product of factors must terminate because the factors are smaller than the composite number itself, and yet each factor is an integer greater than 1. Thus every integer n greater than 1 can be written as a product of primes.

Note (1) The unique expression for the integer $n > 1$ as a product of primes is called the **prime decomposition** or **prime factorisation** of n

(2) If there be n_i prime factors of n , each equal to n_i , where $1 \leq i \leq p$, then n can be written as

$$n = n_1^{a_1} n_2^{a_2} \dots n_p^{a_p}$$

Illustration. The prime factorisation of 100, 216, 9999 are given by

$$100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$$

$$216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 = 2^3 \times 3^3$$

$$9999 = 3 \times 3 \times 11 \times 101 = 3^2 \times 11 \times 101$$

Theorem 2 (Euclid):

The number of prime numbers is infinite [W.B.U.T 2014]

Proof. If possible, let the number of primes be finite and be equal to n . Let them be arranged in order of magnitude as $p_1, p_2, \dots, p_n$.

Then form the number

$$n = 1 + p_1 \cdot p_2 \dots p_n$$

Now no one of p 's is a divisor of n

$\therefore n$ is either a prime $> p_n$ or has a prime $> p_n$ as a factor.

But this contradicts our assumption that p_n is the greatest prime.

Hence the number of primes is infinite.

Theorem 3. If $n > 1$ is a composite integer then there exists a prime p such that $p | n$ and $p \leq \sqrt{n}$

Proof. Since $n > 1$ is a composite integer, n can be expressed as $n = ab$, where $1 < a \leq b < n$. Then

$$a \leq \sqrt{n}, \text{ since if it is not true, } a > \sqrt{n}$$

$$\therefore a \cdot b > \sqrt{n} \cdot \sqrt{n} \Rightarrow ab > n \text{ which is a contradiction}$$

Now by fundamental theorem, either a is a prime or has a prime divisor p

$$\therefore p | a \Rightarrow p | n \text{ and } p \leq \sqrt{n}$$

Note. To check if a given integer n is prime it is sufficient to see that it is not divisible by any prime less than or equal to its square root.

Illustration.

$$\text{Let } n = 19$$

$$\therefore 4 < \sqrt{19} < 5$$

Here 2 and 3 are the prime less than or equal to 4. But 19 is not divisible by 2 and 3

There 19 must be a prime.

Theorem 4 (Division algorithm)

Given any two integers a and b , with $b > 0$, there exist unique integers q and r such that $a = bq + r$, $0 \leq r < b$

Proof. Consider the following infinite sequence of multiples of b :

$$\dots, -2b, -b, 0, b, 2b, \dots, qb, \dots$$

Then, obviously

$$a = qb \text{ or } qb < a < (q+1)b \text{ for some } q$$

In either case we have

$$qb \leq a < (q+1)b \text{ for some } q$$

$$\therefore 0 \leq a - qb < b \quad \dots (1)$$

Let us take $a - bq = r$

$\therefore$ From (1), we get

$$\therefore 0 \leq r < b$$

$$\therefore a = bq + r, 0 \leq r < b$$

To prove the uniqueness of q and r , let us assume

$$a = bq_1 + r_1 \quad \dots (2)$$

where $0 \leq r_1 < b$

$$\text{i.e., } -b < r_1 \leq 0$$

$$\text{and } a = bq_2 + r_2 \quad \dots (3)$$

where $0 \leq r_2 < b$

$\therefore$ (2) and (3) gives

$$bq_1 + r_1 = bq_2 + r_2$$

$$\text{i.e., } (q_1 - q_2)b = r_2 - r_1 \quad \dots (4)$$

Thus $r_2 - r_1$ is an integral multiple of b

But, since $-b < r_1 \leq 0$ and $0 \leq r_2 < b$, so we have

$$-b < r_2 - r_1 < b \quad \dots (5)$$

Hence the only possibility is that $r_2 - r_1$ is zero multiple of b

$$\therefore r_1 = r_2 \text{ and } q_1 = q_2$$

Thus q and r are unique.

Note. (1) When $b \nmid a$, r satisfies the stronger inequalities $0 < r < b$

(2) When b is any integer, the above result can be written as

$$a = bq + r, 0 \leq r < |b|$$

(3) Here q and r are called quotient and remainder respectively

(4) bq is the largest multiple of b which does not exceed a

Illustration

(i) If $a = 57$, $b = 9$, then $57 = 6 \times 9 + 3$, $0 < 3 < 9$

$$\therefore q = 6, r = 3$$

(ii) If $a = -57$, $b = 9$, then

$$-57 = (-7) \times 9 + 6, 0 < 6 < 9$$

$$\therefore q = -7, r = 6$$

(iii) If $a = 57$, $b = -9$, then

$$57 = (-9)(-6) + 3, 0 < 3 < |-9|$$

$$\therefore q = -6, r = 3$$

(iv) If $a = -57$, $b = -9$, then

$$-57 = (-9) \times 7 + 6, 0 < 6 < |-9|$$

$$\therefore q = 7, r = 6.$$

Theorem 5. Given any two integers a and b with $b > 0$, there exist integers Q and R such that $a = bQ + R$, $0 \leq R < b$ (1)

Proof: We have by Division algorithm, for any two integers a and b , $b > 0$, there exist unique q and r such that

$$a = bq + r, 0 \leq r < b \quad \dots (1)$$

We now consider the following cases:

case (i) Let $r < \frac{b}{2}$

Taking $q=Q$ and $r=R$ is (1), we have

$$a = bQ + R, \quad 0 \leq R < \frac{b}{2}$$

case (ii) Let $r > \frac{b}{2}$

Then, from (1)

$$a = bq + r$$

$$= b(q+1) + r - b$$

$$= b(q+1) - (b-r)$$

Taking $q+1=Q$ and $b-r=R$, we have $a=bQ-R$, and

$$R = b-r < b - \frac{b}{2} = \frac{b}{2}$$

$$\therefore a = bQ - R, \quad 0 \leq R < \frac{b}{2}$$

Combining case (i) and case (ii) we have

$$a = bQ \pm R, \quad 0 \leq R < \frac{b}{2}$$

case (iii) Let $r = \frac{b}{2}$

Then, from (1)

$$a = bQ + R, \quad \text{where } q=Q \text{ and } r=R=\frac{b}{2}$$

Again, from (1), we have

$$a = bq + r = b(q+1) - (b-r) = bQ + R$$

where $q+1=Q$, $-(b-r)=R$ i.e. $R = -b + \frac{b}{2} = -\frac{b}{2}$

Thus in this case case Q and R are not unique.

Corollary 1.

Given any two integers a and b , $b > 0$, there exist integers Q and R such that

$$a = bQ + R, \quad 0 \leq |R| \leq \frac{b}{2}$$

Note (i) Here Q and R are unique except when $R = \frac{b}{2}$. In this case i.e. for $R = \frac{b}{2}$, R is called **minimal remainder** i.e. the absolutely least remainder of a w.r.t b .

(ii) When $r < \frac{b}{2}$, the minimal remainder is $R=r$

(iii) When $r > \frac{b}{2}$, the minimal remainder is $R=r-b$

Illustration.

If $a=52$, $b=8$,

then $52 = 6 \times 8 + 4$ (of the form $a=bQ+R$, $Q=6$, $R=4$)

Also $52 = 7 \times 8 - 4$ (of the form $a=bQ-R$, $Q=7$, $R=-4$)

Thus Q and R are not unique as $R = \frac{b}{2} = 4$

Theorem 6. Every integer is of the form

(i) $3k$ or $3k \pm 1$

(ii) $4k$ or $4k \pm 1$ or $4k \pm 2$

(iii) $5k$ or $5k \pm 1$ or $5k \pm 2$

(iv) $6k$ or $6k \pm 1$ or $6k \pm 2$ or $6k \pm 3$

Proof. From corollary 1, we can conclude that any integer a is of the form

$$a = bk + R, \quad 0 \leq |R| \leq \frac{b}{2}, \quad b, k, R \in \mathbb{Z}$$

(i) When $b=3$,

$$0 \leq |R| \leq \frac{3}{2} = 1.5$$

$$\therefore R = 0, \pm 1$$

Therefore, $a = 3k$ or $3k \pm 1$

(ii) When $b=4$,

$$0 \leq |R| \leq \frac{4}{2} = 2$$

$$\therefore R = 0, 0 \pm 1, \pm 2$$

Therefore, $a = 4k$ or $4k \pm 1$ or $4k \pm 2$

(iii) When $b = 5$,

$$0 \leq |R| \leq \frac{5}{2} = 2.5$$

$$\therefore R = 0, \pm 1, \pm 2$$

Therefore, $a = 5k$ or $5k \pm 1$ or $5k \pm 2$

(iv) When $b = 6$,

$$0 \leq |R| \leq \frac{6}{2} = 3$$

$$\therefore R = 0, \pm 1, \pm 2, \pm 3$$

Therefore, $a = 6k$ or $6k \pm 1$ or $6k \pm 2$ or $6k \pm 3$

Illustrative Examples

Ex.1. Show that square of any odd integer is of the form

$$8k+1, \quad k \in \mathbb{Z}^+$$

Solution. Any odd integer is of the form $2n \pm 1 \quad \forall n \in \mathbb{Z}$

$$\begin{aligned} \therefore (2n \pm 1)^2 &= 4n^2 \pm 4n + 1 \\ &= 4n(n \pm 1) + 1 \end{aligned}$$

But, for every value of n , $n(n \pm 1)$ is multiple of 2.

$$\text{So } n(n \pm 1) = 2k, k \in \mathbb{Z}^+$$

$$\text{Thus } (2n \pm 1)^2 = 4 \cdot 2k + 1 = 8k + 1, k \in \mathbb{Z}^+ \quad \checkmark$$

Ex.2. Show that every square integer is of the form

$$9k \text{ or } 3k+1, k \in \mathbb{Z}^+$$

Solution. Every integer is of the form

$$3a, 3a \pm 1 \quad (\text{by Th. 6 (i)}) \text{ for some } a \in \mathbb{Z}$$

$\therefore$ Square of these numbers are of the form:

$$(3a)^2 = 9a^2 = 9k, \text{ where } k = a^2 \in \mathbb{Z}^+$$

$$(3a \pm 1)^2 = 9a^2 \pm 6a + 1$$

$$= 3(3a^2 \pm 2a) + 1$$

$$= 3k + 1 \text{ where } k = 3a^2 \pm 2a \in \mathbb{Z}^+ \quad \forall a \in \mathbb{Z}$$

Hence every square number is of the form

$$9k \text{ or } 3k+1, k \in \mathbb{Z}^+$$

Ex.3. Show that cube of any integer is of the form $9k, 9k \pm 1$, $k \in \mathbb{Z}$

Solution. Every integer is of the form

$$3a, 3a \pm 1 \quad [\text{by Th. 6 (i)}] \text{ for some } a \in \mathbb{Z}$$

Cube of these numbers are of the form

$$(3a)^3 = 27a^3 = 9k, \text{ where } k = 3a^3 \in \mathbb{Z}$$

$$(3a+1)^3 = 27a^3 + 27a^2 + 9a + 1$$

$$= 9(3a^3 + 3a^2 + a) + 1$$

$$= 9k + 1, \text{ where } k = 3a^3 + 3a^2 + a \in \mathbb{Z}$$

$$(3a-1)^3 = 27a^3 - 27a^2 + 9a - 1$$

$$= 9(3a^3 - 3a^2 + a) - 1$$

$$= 9k - 1, \text{ where } k = 3a^3 - 3a^2 + a \in \mathbb{Z}$$

Hence cube of any integer is of the form

$$9k \text{ or } 9k \pm 1$$

Note. Since $(3a-1)^3 = 27a^3 - 27a^2 + 9a - 1$

$$= 9(3a^3 - 3a^2 + a - 1) + 8$$

$$= 9k + 8, \text{ where } k = 3a^3 - 3a^2 + a - 1 \in \mathbb{Z}$$

Thus cube of any integer is of the form

$$9k, 9k+1, 9k+8$$

Ex.4. Show that one of every three consecutive integers is divisible by 3.

Solution. Let $a, a+1, a+2$ be any three consecutive integers.

Again any integer is of the form [By Th. 6(i)]

$$3k, 3k+1, 3k-1, k \in \mathbb{Z}$$

$\therefore$ If $a = 3k$, then a is divisible by 3

If $a = 3k+1$, then $a+2 = 3k+1+2 = 3(k+1)$

$\therefore a+2$ is divisible by 3.

If $a = 3k-1$, then $a+1 = 3k-1+1 = 3k$

$\therefore a+1$ is divisible by 3.

Hence one of every three consecutive integers is divisible by 3

Ex.5(a) Show that the product of any three consecutive integers is divisible by 3!

Solution. Let $a, a+1, a+2$ be any three consecutive positive integers. We shall prove that $a(a+1)(a+2)$, product of three consecutive integer, is divisible by 3!.

For $a=1$, $a(a+1)(a+2) = 1.2.3 = 3!$

$\therefore a(a+1)(a+2)$ is divisible by 3! when $a=1$

let the result is true for $a=k$. Then $k(k+1)(k+2)$ is divisible by 3!

Now for $a=k+1$,

$$a(a+1)(a+2) = (k+1)(k+2)(k+3)$$

$$= k(k+1)(k+2) + 3(k+1)(k+2)$$

Now $(k+1)(k+2)$ is divisible by 2 for all $k \in \mathbb{Z}^+$.

$\therefore 3(k+1)(k+2)$ is divisible by 3! (by assumption)

Hence $a(a+1)(a+2)$ is divisible by 3! when $a=k+1$.

Thus $a(a+1)(a+2)$ is divisible by 3! for $a=k+1$ whenever it is divisible 3! for $a=k$. But $a(a+1)(a+2)$ is divisible 3! for $a=1$.

Hence by the Principle of Mathematical induction $a(a+1)(a+2)$ is divisible by 3! for all $a \in \mathbb{Z}^+$.

Since for any integers a, b , we have

$a|b \Leftrightarrow a|(-b)$, so we can unclude that $a(a+1)(a+2)$ is divisible by 3! $\forall a \in \mathbb{Z}^+$.

Hence the product of any three consecutive integers is divisible by 3!

Alternative solution (without induction).

We have

$$\begin{aligned} \frac{a(a+1)(a+2)}{3!} &= \frac{1.2.3 \dots (a-1).a(a+1)(a+2)}{1.2.3 \dots (a-1)3!} \\ &= \frac{(a+2)!}{(a-1)!3!} = {}^{a+2}C_3 = \text{a positive integer, } \forall a \in \mathbb{Z}^+ \end{aligned}$$

Hence $a(a+1)(a+2)$ is divisible by 3!

$$\forall a \in \mathbb{Z}^+$$

Ex.5(b) Show that the product of any r consecutive natural number is divisible by $r!$ [W.B.U.T 2014]

Solution. Let a be a natural number. Now we have to show that $(a+1)(a+2) \dots (a+r)$ is divisible by $r!$

We have

$$\begin{aligned} \frac{(a+1)(a+2) \dots (a+r)}{r!} &= \frac{1.2.3 \dots a.(a+1)(a+2) \dots (a+r)}{1.2.3 \dots a.r!} \\ &= \frac{(a+r)!}{a!r!} = {}^{a+r}C_r = \text{a positive integer} \end{aligned}$$

Hence $(a+1)(a+2) \dots (a+r)$ is divisible by $r!$

Ex.6. Show that the number of the form $\frac{p(p^4+4)}{5}$ is an integer where $p \geq 1$ is an integer.

Solution. Every integer is of the form,

$$5a \text{ or } 5a \pm 1 \text{ or } 5a \pm 2 \text{ [By Th 6(iii)] where } a \in \mathbb{Z}.$$

Let $p = 5a$.

$$\text{Then } \frac{p(p^4+4)}{5} = \frac{5a(625a^4+4)}{5} = a(625a^4+4)$$

which is an integer as $a \in \mathbb{Z}$

Again let $p = 5a \pm 1$

$$\begin{aligned} \text{Then } \frac{p(p^4+4)}{5} &= \frac{(5a \pm 1)(625a^4 \pm 500a^3 + 150a^2 \pm 20a + 5)}{5} \\ &= (5a \pm 1)(125a^4 \pm 100a^3 + 30a^2 \pm 4a + 1) \end{aligned}$$

which is also an integer as $a \in \mathbb{Z}$.

Next let $p = 5a \pm 2$

$$\begin{aligned} \text{Then } \frac{p(p^4+4)}{5} &= \frac{(5a \pm 2)(625a^4 \pm 1000a^3 + 600a^2 \pm 160a + 20)}{5} \\ &= (5a \pm 2)(125a^4 \pm 200a^3 + 120a^2 \pm 32a + 4) \end{aligned}$$

which is an integer as $a \in \mathbb{Z}$

Hence the result.

Ex. 7. Find the minimal remainder of 345 with respect to

- (i) 31 (ii) 27

Solution. (i) Here $a = 345$, $b = 31$.

$$\therefore 345 = 31 \times 11 + 4$$

$$\therefore \text{Remainder, } r = 4$$

Since $4 < \frac{31}{2}$, the minimal remainder of 345 w.r.t 31 is

$$R = 4 \quad [\text{By note (ii)}]$$

(ii) Here $a = 345$, $b = 27$

$$\therefore 345 = 27 \times 12 + 21$$

$$\therefore r = 21$$

Since $21 > \frac{27}{2}$, the minimal remainder of 345 w.r.t 27 is

$$R = 21 - 27 = -6 \quad [\text{By note (iii)}]$$

Ex. 8. If $(a-s)|(ab+st)$, then show that $(a-s)|(at+bs)$ where $a, b, s, t \in \mathbb{Z}$

Solution. Since $(a-s)|(ab+st)$, we have

$$(ab+st) = p(a-s) \quad \text{where } p \in \mathbb{Z}$$

$$\text{Now } (ab+st) - (at+bs) = (a-s)(b-t)$$

$$\therefore p(a-s) - (at+bs) = (a-s)(b-t)$$

$$\therefore (at+bs) = (p-b+t)(a-s)$$

$$= q(a-s) \quad \text{where } q = p-b+t \in \mathbb{Z} \text{ as } p, b, t \in \mathbb{Z}$$

$$\therefore (a-s)|(at+bs)$$

Ex. 9. Show that $x^n - y^n$ is divisible by $x - y$ where n is any positive integer greater than one.

Solution. Since $x^2 - y^2 = (x-y)(x+y)$, so the result is true for $n = 2$. Let us assume that the result is true for $n = k$ for some $k > 2$. This gives $x^k - y^k = m(x-y)$ where m is an integer.

$$\text{Now } x^{k+1} - y^{k+1} = x(x^k - y^k) + y^k(x-y)$$

$$= x.m(x-y) + y^k(x-y) = (x-y)(mx + y^k)$$

$$= n(x-y) \quad \text{where } n = mx + y^k \text{ is an integer}$$

Hence $x^n - y^n$ is divisible by $x - y$ for $n = k + 1$ whenever $x^n - y^n$ is divisible by $x - y$ for $n = k$. But $x^n - y^n$ is divisible by $x - y$ for $n = 2$. Hence by the Principle of Mathematical induction $x^n - y^n$ is divisible by $x - y$ where n is any positive integer greater than one.

Note. As $x^n - y^n$ is divisible by $x - y$, so we can write

$$x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + y^{n-1})$$

Ex.10. Show that $2903^n - 803^n - 464^n + 261^n$ is divisible 1897 where n is any natural number.

Solution. Let $f(n) = 2903^n - 803^n - 464^n + 261^n$

$$= (2903^n - 803^n) - (464^n - 261^n)$$

Now $2903^n - 803^n$ is divisible by $2903 - 803 = 2100$ and $464^n - 261^n$ is divisible by $464 - 261 = 203$ [by Ex-9].

Again since 2100 and 203 are both divisible by 7, it follows that $f(n)$ is divisible by 7.

We can write $f(n)$ as $f(n) = (2903^n - 464^n) - (803^n - 261^n)$

where $2903^n - 464^n$ is divisible by $2903 - 464 = 2439$ and

$803^n - 261^n$ is divisible by $803 - 261 = 542$. But 2439 and 542 are both divisible by 271. Thus $f(n)$ is divisible by 271.

As 271 and 7 are prime to each other and also they are both primes, so $f(n)$ is divisible by $271 \times 7 = 1897$.

Ex.11 Show that $x^{n+1} - (x-1)n - x$ is divisible by $(x-1)^2$, x being an integer.

Solution. We have

$$\begin{aligned} x^{n+1} - (x-1)n - x &= x(x^n - 1) - (x-1)n \\ &= x(x-1)(x^{n-1} + x^{n-2} + \dots + x + 1) - (x-1)n \\ &= (x-1)\{(x^n + x^{n-1} + \dots + x^2 + x) - n\} \\ &= (x-1)\{(x^n - 1) + (x^{n-1} - 1) + \dots + (x^2 - 1) + (x - 1)\} \\ &= (x-1)^2 \{(x^{n-1} + x^{n-2} + \dots + x + 1) + (x^{n-2} + x^{n-3} + \dots + x + 1) \\ &\quad \dots + (x + 1) + 1\} \\ &= m(x-1)^2 \text{ where } m \text{ is an integer.} \end{aligned}$$

Thus $x^{n+1} - (x-1)n - x$ is divisible by $(x-1)^2$

Ex.12. Show that $2^n + 1$ or $2^n - 1$ is divisible by 3 according as n is odd or even

Solution. Let $n = 2m$, $m \in \mathbb{Z}$

Then $2^n - 1$

$$= 2^{2m} - 1 = 4^m - 1$$

$$= (4-1)(4^{m-1} + 4^{m-2} + \dots + 4 + 1)$$

$$= 3k \text{ when } k = 4^{m-1} + 4^{m-2} + \dots + 4 + 1 \text{ is an integer.}$$

Thus $2^n - 1$ is divisible by 3 when n is even

Next let $n = 2m+1$, $m \in \mathbb{Z}$

$$\therefore 2^n + 1$$

$$= 2^{2m+1} + 1 = 4^m \cdot 2 + 1$$

$$= (1+3k)2 + 1 \quad [\because 4^m - 1 = 3k \Rightarrow 4^m = 1 + 3k]$$

$$= 3 + 6k = 3(1+2k).$$

$\therefore 2^n + 1$ is divisible by 3 when n is odd.

Ex.13. If a is an integer not divisible by 2 or 3, then prove that

$$24 \mid (a^2 + 23)$$

Solution. Now $24 \mid (a^2 + 23)$

$$\Rightarrow a^2 + 23 = 24p \text{ for some } p \in \mathbb{Z}$$

$$\Rightarrow a^2 = 24p - 23$$

$$\Rightarrow a^2 = 24(p-1) + 1$$

$$\Rightarrow a^2 = 2m+1 \text{ or } 3n+1 \text{ when } m = 12(p-1) \text{ and } n = 8(p-1)$$

$$\Rightarrow a \neq 2m \text{ or } 3n$$

$$\Rightarrow 2 \nmid a \text{ or } 3 \nmid a$$

1.3.6. Greatest Common Divisor

Definition-1 The integer $d (\neq 0)$ is said to be the common divisor of a and b if $d \mid a$ and $d \mid b$ i.e., if d divides both a and b .

Since there is only a finite number of divisors of any non-zero integers, there is only a finite number of common divisors of a and b , except in the case $a = b = 0$. If at least one of a and b is not 0, the largest of all common divisors is called the **greatest common divisor** of a and b and is denoted by $\gcd(a, b)$ or simply (a, b) . Obviously $\gcd(a, b) \geq 1$.

If $\gcd(a, b) = 1$, then a and b are said to be **relatively prime** or **coprime** or each said to be prime to the other.

If $\gcd(a_1, a_2, \dots, a_n) = 1$, then the integers $a_1, a_2, \dots, a_n$ are said to be pairwise relatively prime.

Definition-2 The greatest common divisor of any n number of given integers, $a_1, a_2, \dots, a_n$, $a_i \neq 0, \forall i = 1, 2, \dots, n$, denoted by $\gcd(a_1, a_2, \dots, a_n)$ or, $(a_1, a_2, \dots, a_n)$ and is defined as the positive integer d , satisfies the following properties

$$(i) d | a_i, \forall i = 1, 2, \dots, n$$

$$(ii) \text{ if } c | a_i, i = 1, 2, \dots, n, \text{ then } c | d$$

Illustration.

(i) The divisors of 8 are $\pm 1, \pm 2, \pm 4, \pm 8$; the divisors of 36 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 9, \pm 12, \pm 18, \pm 36$

Thus the common divisors of 8 and 36 are $\pm 1, \pm 2, \pm 4$

Hence the greatest common divisor is 4.

$$\therefore \gcd(8, 36) = 4$$

1.3.7. Euclidean algorithm for finding GCD

Theorem.7: Let a and b ($a > b$) be any two integers. If r_1 is the remainder when a is divided by b , r_2 is the remainder when b is divided by r_1 , r_3 is the remainder when r_1 is divided by r_2 and so on. Thus if $r_{p+1} = 0$, then the last non-zero remainder r_p is the $\gcd(a, b)$.

Proof. Let $a = qb + r, 0 \leq r < b$

We now prove that $\gcd(a, b) = \gcd(b, r)$

For that, let $d_1 = \gcd(a, b)$... (1)

and $d_2 = \gcd(b, r)$... (2)

$\therefore$ From (2),

$$d_2 | b \text{ and } d_2 | r$$

$$\therefore d_2 | (qb + r) \text{ i.e., } d_2 | a$$

Thus d_2 is a common divisor of a and b . Since d_1 is $\gcd(a, b)$, we must have

$$d_2 \leq d_1 \quad \dots (3)$$

Again from (1),

$$d_1 | a \text{ and } d_1 | b$$

$$\therefore d_1 | (a - qb) \text{ i.e., } d_1 | r$$

Thus d_1 is a common divisor of b and r . Since $d_2 = \gcd(b, r)$, we must have

$$d_1 \leq d_2 \quad \dots (4)$$

$\therefore$ From (3) and (4), we get

$$d_1 = d_2 \quad \dots (5)$$

$$\text{i.e., } \gcd(a, b) = \gcd(b, r) \text{ where } a = qb + r$$

Now, since r_1 is the remainder when a is divided by b , we have $a = q_1b + r_1, 0 \leq r_1 < b$.

Similarly by the given data,

$$b = q_2r_1 + r_2, 0 \leq r_2 < r_1$$

$$r_1 = q_3r_2 + r_3, 0 \leq r_3 < r_2$$

$$\dots \dots \dots$$

$$r_{p-1} = q_{p+1}r_p + r_{p+1}, 0 \leq r_{p+1} < r_p$$

Since $r_1, r_2, r_3, \dots$ form a decreasing set of non-negative integers, there must exist an r_{p+1} equal to zero.

Using (5) we have

$$\begin{aligned}\gcd(a, b) &= \gcd(b, r_1) = \gcd(r_1, r_2) = \dots \\ &= \gcd(r_{p-1}, r_p) = \gcd(r_p, 0) = r_p\end{aligned}$$

Hence $\gcd(a, b) = r_p$ which is the last non-zero remainder.

Illustration. We now find $\gcd(1120, 128)$

By division algorithm,

$$1120 = 8 \times 128 + 96$$

$$128 = 1 \times 96 + 32$$

$$96 = 3 \times 32 + 0$$

Since the last non-zero remainder is 32, so

$$\gcd(1120, 128) = 32$$

Theorem 8. Let a and b ($a > b$) be any two integers. Then $\gcd(a, b)$ can be expressed as an integral linear combination of a and b .

i.e., $\gcd(a, b) = ma + nb$, where m and n are integers.

Proof. Using the Euclid's algorithm, we have

$$r_{p-2} = q_p r_{p-1} + r_p, \text{ where } r_p = \gcd(a, b)$$

$$\therefore r_p = r_{p-2} + (-q_p) r_{p-1} \quad \dots (1)$$

$$\therefore r_{p-1} = r_{p-3} + (-q_{p-1}) r_{p-2}$$

$\therefore$ From (1),

$$\begin{aligned}r_p &= r_{p-2} + (-q_p) \{r_{p-3} + (-q_{p-1}) r_{p-2}\} \\ &= r_{p-3} + (-q_p) + (1 + q_{p-1} q_p) r_{p-2}\end{aligned}$$

Then we substitute $r_{p-4} + (-q_{p-2}) r_{p-3}$ for r_{p-3} and continue the process. Finally we will have

$$r_p = \gcd(a, b) = ma + nb \text{ for some integers } m \text{ and } n$$

Illustration. First we consider the steps used to find $\gcd(1120, 128)$ as given below

$$1120 = 8 \times 128 + 96 \quad \dots (1)$$

$$128 = 1 \times 96 + 32 \quad \dots (2)$$

$\therefore$ From (2), we have

$$32 = 128 - 1 \times 96$$

$$= 128 - 1 \times (1120 - 8 \times 128), \text{ by (1)}$$

$$= 9 \times 128 - 1 \times 1120 \quad \dots (3)$$

$$\text{Thus } 32 = \gcd(1120, 128) = (-1)1120 + 9 \times 128$$

$$\therefore m = -1, n = 9$$

Note. (3) can be written as

$$\gcd(1120, 128) = (9 + 1120) \times 128 - (1 + 128) \times 1120$$

$$= 1129 \times 128 - 129 \times 1120$$

$$= (-129)1120 + 1129 \times 128$$

$$\therefore m = -129, n = 1129$$

Thus m and n are not unique.

Hence the expression of $\gcd(a, b)$ in the form $ma + nb$ is not unique.

Alternative definition of GCD

Let the prime decomposition of two integers a and b be

$$a = n_1^{a_1} n_2^{a_2} n_3^{a_3} \dots n_r^{a_r}$$

and

$$b = n_1^{b_1} n_2^{b_2} n_3^{b_3} \dots n_r^{b_r}$$

where each exponent is a non-negative integer and all primes occurring in the prime decomposition of either a or b are included in both decomposition, with zero exponent if necessary. Then

$$\gcd(a, b) = n_1^{\min(a_1, b_1)} n_2^{\min(a_2, b_2)} \dots n_r^{\min(a_r, b_r)}$$

where $\min(a_i, b_i)$ means the minimum of two numbers a_i and b_i

Illustration.

We have

$$20 = 2^2 \cdot 5^1 \cdot 7^0$$

$$350 = 2^1 \cdot 5^2 \cdot 7^1$$

$$\therefore \gcd(20, 350) = 2^{\min(2,1)} \cdot 5^{\min(1,2)} \cdot 7^{\min(0,1)}$$

$$= 2^1 \cdot 5^1 \cdot 7^0$$

$$= 10$$

1.3.8 . Relatively prime integers.

Definition. Two integers a and b are said to be **relatively prime** or **coprime** or **prime to each other** if $\gcd(a, b) = 1$.

Definition. The integers $a_1, a_2, \dots, a_n$ are said to be **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$.

Illustration.

(i) The divisors of 15 are $\pm 1, \pm 3, \pm 5, \pm 15$; the divisors of 44 are $\pm 1, \pm 2, \pm 4, \pm 11$.

$$\text{Hence } \gcd(15, 44) = 1.$$

$\therefore$ 15 and 44 are relatively prime

(ii) Consider the integers 8, 17, 35. Since $\gcd(8, 17) = 1$, $\gcd(8, 35) = 1$ and $\gcd(17, 35) = 1$, so the integers 8, 17, 35 are pairwise relatively prime.

Theorem.9. If a and b be integers, not both zero, then a and b are prime to each other if and only if there exist integers u and v such that $au + bv = 1$.

In other words $\gcd(a, b) = 1 \Leftrightarrow au + bv = 1$ for some integers u and v [W.B.U.T.2013]

Proof. First, let a and b are prime to each other.

$$\therefore \gcd(a, b) = 1 \quad (1)$$

We have, from Theorem-6,

$$\gcd(a, b) = ua + vb \text{ for some integers } u \text{ and } v.$$

$\therefore$ from (1),

$$au + bv = 1$$

Next let there exist integers u and v such that

$$au + bv = 1$$

Let $d = \gcd(a, b)$

$$\therefore d|a \text{ and } d|b$$

$\therefore$ There exist integers u and v such that

$$d|au + bv, \text{ by Theorem-1(iv)}$$

$$d|1$$

$$\therefore d = 1, \text{ since } d > 0$$

$$\text{Hence } \gcd(a, b) = 1$$

1.3.9. Some properties of GCD

$$\text{A. } \gcd(a, b) = \gcd(b, a) \quad \forall a, b \in \mathbb{Z} \quad [\text{commutative property}]$$

Proof. Left as an exercise.

$$\text{B. } \gcd(\gcd(a, b), c) = \gcd(a, \gcd(b, c)), \quad \forall a, b, c \in \mathbb{Z} \quad [\text{Associative property}].$$

Proof. Let $\gcd(a, b) = d_1$

$$\text{Then } \gcd(\gcd(a, b), c)$$

$$= \gcd(d_1, c)$$

$$= d_2, \text{ say.}$$

$$\text{Now } \gcd(d_1, c) = d_2 \Rightarrow d_2 | d_1, d_2 | c$$

$$\text{and } \gcd(a, b) = d_1 \Rightarrow d_1 | a, d_1 | b$$

Thus we have

$$d_2 | a, d_2 | b, d_2 | c$$

Next, if $d|d_1, d|c$, then $d|d_2$ as $\gcd(d_1, c) = d_2$

Also $d|d_1$ and $d_1|a \Rightarrow d|a$

Similarly $d|b$

Thus if $d|a, d|b$ and $d|c$ then $d|d_2$

$\therefore \gcd(a, b, c) = d_2$, by Definition-2

Similarly we can show that

$$\gcd(a, \gcd(b, c)) = \gcd(a, b, c) = d_2$$

$$\text{Hence } \gcd(\gcd(a, b), c) = \gcd(a, \gcd(b, c))$$

C. For any positive integer m ,

$$\gcd(ma, mb) = m \gcd(a, b) \quad [\text{Distributive property}]$$

[W.B.U.T.2014]

Proof. Let $d = \gcd(a, b)$

By theorem 5, there exists integers x and y that

$$xa + yb = d$$

$$\therefore m(xa + yb) = md$$

$$\text{i.e., } x(ma) + y(mb) = md$$

$$\therefore \gcd(ma, mb) = m \gcd(a, b)$$

Note. When m is any integer, the result becomes

$$\gcd(ma, mb) = |m| \gcd(a, b)$$

D. If $c|ab$ and b, c are coprime, then $c|a$

Proof. Since b, c are coprime, so

$$\gcd(b, c) = 1.$$

$\therefore$ By theorem 5, there exist integers m and n such that

$$mb + nc = \gcd(b, c)$$

$$\text{i.e., } mb + nc = 1$$

$$\text{i.e., } a(mb + nc) = a$$

$$\text{i.e., } mab + nac = a \quad \dots (1)$$

Now $c|ab \Rightarrow c|mab$

Also $c|nac$

$$\therefore c|(mab + nac)$$

$$\therefore c|a, \text{ by (1)}$$

E. If a and b are coprime and a and c are coprime, then a and bc are coprime i.e., $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$ then $\gcd(a, bc) = 1$

Proof. Since a, b are coprime, so

$$\gcd(a, b) = 1$$

$\therefore$ There exists integers m and n such that

$$ma + nb = 1 \quad \dots (2)$$

Similarly, since a and c are coprime, there exists integers x and y such that

$$xa + yc = 1 \quad \dots (3)$$

$\therefore$ From (2) and (3), we get

$$(ma + nb)(xa + yc) = 1$$

$$\text{i.e., } (mxa + myc + nxb) + (ny)bc = 1$$

which is of the form

$$pa + qbc = 1, \text{ where } p = mxa + myc + nxb, q = ny$$

are integers

$$\text{Hence } \gcd(a, bc) = 1$$

$\therefore a$ and bc are coprime

F. If $d|a$ and $d|b$ and $d > 0$, then

$$\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = \frac{1}{d} \gcd(a, b)$$

Proof. Let $\gcd(a, b) = k$

$\therefore$ By theorem 5, there exists integers m and n such that

$$ma + nb = k$$

$$\therefore \frac{1}{d}(ma + nb) = \frac{k}{d}$$

$$\text{i.e. } m\left(\frac{a}{d}\right) + n\left(\frac{b}{d}\right) = \frac{1}{d} \cdot k$$

Since $d \mid a$ and $d \mid b$, $\frac{a}{d}$ and $\frac{b}{d}$ are integers.

$$\text{Hence } \gcd\left(\frac{a}{d}, \frac{b}{d}\right) = \frac{1}{d} \cdot \gcd(a, b)$$

Note. If $\gcd(a, b) = d$, then $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$

$\therefore \frac{a}{d}$ and $\frac{b}{d}$ are prime to each other.

G. If $\gcd(a, b) = 1$, then for any integer x ,

$$\gcd(ax, b) = \gcd(x, b)$$

Proof. Since $\gcd(a, b) = 1$, by theorem 5, there exists integers m, n such that

$$ma + nb = 1 \quad \dots (1)$$

Let $\gcd(ax, b) = d$.

Then there exists integers p, q such that

$$pax + qb = d \quad \dots (2)$$

$\therefore$ From (1) and (2)

$$(ma + nb)(pax + qb) = 1 \cdot d$$

$$\text{i.e., } (mpa^2)x + (maq + npax + nqb)b = d$$

$$\text{i.e., } rx + sb = d \text{ where } r = mpa^2, s = maq + npax + nqb$$

are integers.

Hence $\gcd(x, b) = d$

$$\therefore \gcd(ax, b) = \gcd(x, b).$$

H. If $a_1, a_2, \dots, a_n$ are relatively prime to b , then their product $a_1 a_2 \dots a_n$ is also prime to b

Proof. Since a_1 is relatively prime to b , so

$$\gcd(a_1, b) = 1$$

$\therefore$ By property G,

$$\begin{aligned} \gcd(a_1 a_2, b) &= \gcd(a_2, b) \\ &= 1 \quad [\because a_2, b \text{ are relatively prime}] \end{aligned}$$

Again by property G,

$$\begin{aligned} \gcd(a_1 a_2 a_3, b) &= \gcd(a_3, b) \\ &= 1 \quad [\because a_3, b \text{ are relatively prime}] \end{aligned}$$

Proceeding in this way, we get

$$\gcd(a_1 a_2 a_3 \dots a_n, b) = 1.$$

Thus $a_1 a_2 a_3 \dots a_n$ is prime to b .

1.3.10. Least Common Multiple.

Definition. Let a and b be positive integers. Then the smallest positive integer that is divisible by both a and b is called the least common multiple of a and b and is denoted by $\text{lcm}(a, b)$ or $[a, b]$

Note. $\text{lcm}(a, b)$ is always positive even if either or both a and b are negative.

Illustration.

$$\text{lcm}(8, 20) = \text{lcm}(-8, 20) = \text{lcm}(-8, -20) = 40$$

Alternative definition of lcm

Let the prime decomposition of two integers a and b be

$$a = n_1^{a_1} n_2^{a_2} \dots n_p^{a_p}$$

$$\text{and } b = n_1^{b_1} n_2^{b_2} \dots n_p^{b_p}$$

where each exponent is a non-negative integer and all primes occurring in the prime decomposition of either a or b are included in both decomposition, with zero exponent if necessary. Then

$$\text{lcm}(a, b) = n_1^{\max(a_1, b_1)} n_2^{\max(a_2, b_2)} \dots n_p^{\max(a_p, b_p)}$$

where $\max(a_i, b_i)$ means the maximum of two numbers a_i and b_i

Illustration.

We have $8 = 2^3 \times 5^0$

$20 = 2^2 \times 5^1$

$\therefore \text{lcm}(8, 20) = 2^{\max(3, 2)} \cdot 5^{\max(0, 1)}$

$= 2^3 \cdot 5^1 = 40$

✓ **Theorem 9** Let a and b be any two positive integers.

Then $\text{gcd}(a, b) \cdot \text{lcm}(a, b) = ab$ ✓

Proof. Let the prime decomposition of a and b be

$a = n_1^{a_1} n_2^{a_2} n_3^{a_3} \dots n_p^{a_p}$

and $b = n_1^{b_1} n_2^{b_2} n_3^{b_3} \dots n_p^{b_p}$

$\therefore \text{gcd}(a, b) = n_1^{\min(a_1, b_1)} n_2^{\min(a_2, b_2)} \dots n_p^{\min(a_p, b_p)}$

$\text{lcm}(a, b) = n_1^{\max(a_1, b_1)} n_2^{\max(a_2, b_2)} \dots n_p^{\max(a_p, b_p)}$

Note that if $\max(a_i, b_i)$ is a_i (or b_i), then $\min(a_i, b_i)$ is b_i (or a_i), $i = 1, 2, \dots, p$

$\therefore \text{gcd}(a, b) \times \text{lcm}(a, b)$

$= n_1^{\{\max(a_1, b_1) + \min(a_1, b_1)\}} \cdot n_2^{\{\max(a_2, b_2) + \min(a_2, b_2)\}}$

$\dots n_p^{\{\max(a_p, b_p) + \min(a_p, b_p)\}}$

$= n_1^{a_1+b_1} n_2^{a_2+b_2} \dots n_p^{a_p+b_p}$

$= (n_1^{a_1} n_2^{a_2} \dots n_p^{a_p}) \cdot (n_1^{b_1} n_2^{b_2} \dots n_p^{b_p})$

$= ab$

Illustration.

Since $\text{lcm}(8, 20) = 40$, so

$\text{gcd}(8, 20) \cdot \text{lcm}(8, 20) = 8 \times 20$ gives

$\text{gcd}(8, 20) = \frac{8 \times 20}{40} = 4$

✓ **Theorem 10.** Show that $(a+b, \text{lcm}(a, b)) = \text{gcd}(a, b)$ for any two non-zero integers a and b

Proof. Let $d = \text{gcd}(a, b)$ and $R = \text{lcm}(a, b)$

Then $d = \text{gcd}(a, b)$

$\Rightarrow d|a$ and $d|b$

Also $R = \text{lcm}(a, b) \Rightarrow a|R$ and $b|R$

$\therefore d|a$ and $a|R \Rightarrow d|R$

Also $d|a, d|b \Rightarrow d|(a+b)$ (1)

We now consider the followings two cases:

Let d be prime.

Then $(a+b, \text{lcm}(a, b))$

$= (a+b, R)$

$= d$ by (1)

$= \text{gcd}(a, b)$

Next let d is not prime. So d is a composite number.

$\therefore d = d_1 d_2$ (say) where $(d_1, d_2) = 1, d_1, d_2 < d$ (or d_2)

Then there exist a positive number d_1 (say) such that.

$d_1|a+b, d_1|R$ and $d_1|d$.

$\therefore (a+b, \text{lcm}(a, b))$

$= (a+b, R) = d = \text{gcd}(a, b)$.

Illustration.

(i) Let $a = 18, b = 54$

$\therefore \text{gcd}(a, b) = 2$, a prime number

Now $a+b = 72, \text{lcm}(a, b) = 162$

$\therefore \text{gcd}(a+b, \text{lcm}(a, b))$

$= \text{gcd}(72, 162) = 2 = \text{gcd}(a, b)$

(ii) Let $a=18$, $b=66$

$\therefore \gcd(a, b) = 6$, a composite number

Now $a+b=84$, $\text{lcm}(a, b)=198$

$$\therefore \gcd(a+b, \text{lcm}(a, b)) = \gcd(84, 198) = 6 = \gcd(a, b)$$

1.3.11 Diophantine Equations.

The general form of a linear Diophantine equation in two variables having integral coefficients is

$$ax + by = c \quad \dots (1)$$

The solution of this equation is trivial unless neither a nor b is zero, so we can suppose $a \neq 0$, $b \neq 0$. Let $d = \gcd(a, b)$. Then theorem 5 shows that there exist integers x_0 and y_0 such that

$$ax_0 + by_0 = d \quad \dots (2)$$

Numerical values for x_0 and y_0 can be obtained conveniently by applying the Euclidean algorithm, art. 2.1.7 to the integers $|a|$ and $|b|$. Now if $d|c$, we get (x_0, y_0) as the particular integral solution of (1). Then all integral solutions of (1) are given by

$$x = x_0 + \frac{b}{d}k, \quad y = y_0 - \frac{a}{d}k$$

where k is any integer.

But, if $d \nmid c$, then $ax + by = c$ clearly has no solutions in integers.

Illustration.

(i) Consider the Diophantine equation $2x + 3y = 4$.

Here $\gcd(2, 3) = 1$ and 4 is divisible by 1. So the equation $2x + 3y = 4$ has integral solution.

$$\text{Now } 1 = 2 \cdot (-1) + 3 \cdot 1$$

$$\therefore 4 = 2 \cdot (-4) + 3 \cdot 4$$

Thus one integral solution is $x_0 = -4$, $y_0 = 4$.

Hence all integral solutions of the given equation are given by

$$x = -4 + \frac{3}{1}k, \quad y = 4 - \frac{2}{1}k \text{ for any integer } k$$

i.e., $x = -4 + 3k$, $y = 4 - 2k$ for any integer k .

(ii) Consider another Diophantine equation

$$14x + 21y = 5$$

Here $\gcd(14, 21) = 7$ and 5 is not divisible by 7. Hence the given equation has no integral solution.

1.3.12. Illustrative Examples.

Ex.1. If $a|c$ and $b|c$ with $\gcd(a, b)=1$, prove that $ab|c$

Solution. Since $a|c$ and $b|c$, there exist integers p and q such that $c = pa$, $c = bq$

Again $\gcd(a, b)=1$, so there exist integer $u, v \in \mathbb{Z}$ such that

$$1 = au + bv$$

$$c = (au)c + (bv)c$$

$$= au.bq + bv.pa$$

$$= ab(uq + vp)$$

$$\Rightarrow ab|c \quad [\because uq + vp \in \mathbb{Z}]$$

Ex. 2. If $d = \gcd(x, y)$, then $\frac{x}{d}$ and $\frac{y}{d}$ are integers prime to each other.

Solution. Since $d = \gcd(x, y)$, there exist integers l and m such that

$$d = lx + my$$

$$\therefore 1 = \frac{x}{d}l + \frac{y}{d}m$$

Since d is the division of x and y , there exist integers u and v

such that $\frac{x}{d} = u$, $\frac{y}{d} = v$

$$\therefore 1 = ul + vm$$

$\therefore u$ and v are integers prime to each other.

Thus $\frac{x}{d}$ and $\frac{y}{d}$ are integers prime to each other.

Ex.3. If $\gcd(a, 4) = 2$ and $\gcd(b, 4) = 2$, proved that

$$\gcd(a+b, 4) = 4$$

Solution. Since $\gcd(a, 4) = 2$, so a is a multiple of 2 but not 4

$$\therefore a = 2p, \text{ for some odd integer } p$$

Similarly, since $\gcd(b, 4) = 2$, we have

$$b = 2q, \text{ for some odd integer } q$$

$$\therefore a+b = 2(p+q)$$

$$= 2 \cdot 2r \text{ where } p+q = 2r, \text{ an even integer,}$$

as, p, q are odd

$$= 4r$$

$$\therefore \gcd(a+b, 4) = \gcd(4r, 4) \text{ where } r \text{ is an integer}$$

$$= 4$$

$$\text{Thus } \gcd(a+b, 4) = 4.$$

Ex.4. If $\gcd(a, b) = 1$, prove that $\gcd(a+b, ab) = 1$

Solution: Since $\gcd(a, b) = 1$, there exist integer $u, v \in \mathbb{Z}$ such that

$$au + bv = 1 \quad (1)$$

which can be written as

$$a(u-v) + (a+b)v = 1$$

$$\therefore \gcd(a+b, a) = 1 \quad [\because u-v, v \in \mathbb{Z}] \quad (2)$$

Again (1) can also be written as

$$(a+b)u + b(v-u) = 1$$

$$\therefore \gcd(a+b, b) = 1 \quad [\because u, v-u \in \mathbb{Z}] \quad (3)$$

$\therefore$ In virtue of property E of 2.1.8, we can conclude from (2) and (3) that

$$\gcd(a+b, ab) = 1$$

Ex. 5. If $\gcd(a, b) = 1$ prove that $\gcd(a+b, a-b) = 1$ or 2

Solution Let $\gcd(a+b, a-b) = d$

$$\therefore a+b = k_1 d \quad \dots (1)$$

$$\text{and } a-b = k_2 d \quad \dots (2)$$

where k_1, k_2 are integers

From (1) and (2), we get

$$2a = (k_1 + k_2)d \text{ and } 2b = (k_1 - k_2)d$$

$$\therefore d \text{ divides } 2a \text{ and } 2b$$

$$\therefore d \leq \gcd(2a, 2b) = 2\gcd(a, b), \text{ by property (C) of at 2.1.8}$$

$$= 2 \cdot 1 \quad [\because \gcd(a, b) = 1]$$

$$= 2.$$

Hence $d = 1$ or 2

$$\therefore \gcd(a+b, a-b) = 1 \text{ or } 2$$

✓ Ex.6. If $\gcd(a, b) = 1$, prove that $\gcd(a^2, b^2) = 1$

Solution Since $\gcd(a, b) = 1$, there exist integers $u, v \in \mathbb{Z}$ such that

$$au + bv = 1,$$

$$\therefore au = 1 - bv$$

$$\Rightarrow a^2 u^2 = 1 - 2bv + b^2 v^2$$

$$\Rightarrow a^2 u^2 + b(2v - bv^2) = 1$$

$$\Rightarrow pa^2 + bq = 1$$

$$\text{where } p = u^2 \in \mathbb{Z} \text{ and } q = 2v - bv^2 \in \mathbb{Z}$$

$$\Rightarrow bq = 1 - pa^2$$

$$\Rightarrow b^2 q^2 = 1 - 2pa^2 + p^2 a^4$$

$$\Rightarrow a^2 (2p - p^2 a^2) + b^2 q^2 = 1$$

$$\therefore \gcd(a^2, b^2) = 1 \quad [\because 2p - p^2 a^2 \in \mathbb{Z} \text{ and } q^2 \in \mathbb{Z}]$$

Ex.7. If $\gcd(a, b) = k$, prove that $\gcd(a^2, b^2) = k^2$

Solution: As $\gcd(a, b) = k$, so we have

$$k|a \text{ and } k|b$$

Thus there exist integers p, q prime to each other such that

$$a = pk, \quad b = qk$$

$$\therefore a^2 = p^2 k^2, \quad b^2 = q^2 k^2$$

$$\therefore k^2 \text{ is a common divisor of } a^2 \text{ and } b^2$$

Let us assume that $\gcd(a^2, b^2) = k^2 r$ where $r \in \mathbb{Z}^+$

$$\therefore k^2 r | a^2 \text{ and } k^2 r | b^2$$

$$\Rightarrow k^2 r | p^2 k^2 \text{ and } k^2 r | q^2 k^2, \quad \text{by (1)}$$

$$\Rightarrow r | p^2 \text{ and } r | q^2 \quad (2)$$

$$\text{But } \gcd(p, q) = 1$$

$$\therefore \gcd(p^2, q^2) = 1 \quad [\text{by Ex.6}] \quad (3)$$

In virtue of (2) and (3), we get

$$r = 1$$

$$\text{Hence } \gcd(a^2, b^2) = k^2$$

Ex.8. If $\gcd(a, b) = 1$, prove that for any integer p , $\gcd(a + pb, b) = 1$

Solution. Let $\gcd(a + pb, b) = k$, where $k \geq 1$

If $k = 1$, the result follows.

If $k > 1$, then

$$\gcd(a + pb, b) = k \quad (>1)$$

$$\Rightarrow k | a + pb, \quad k | b$$

$$\Rightarrow k | (a + pb) + b(-p), \quad k | b \text{ by linearity property of divisibility}$$

$$\Rightarrow k | a \text{ and } k | b$$

$$\Rightarrow \gcd(a, b) \neq 1, \text{ a contradiction}$$

$$\text{Hence } \gcd(a + pb, b) = 1$$

Note $\gcd(a, b) = 1 \Rightarrow \gcd(a + b, b) = 1$

Ex.9. If $\gcd(a, b) = 1$, prove that

$$\gcd(a + b, a^2 - ab + b^2) = 1 \text{ or } 3 \quad [W.B.U.T.2014]$$

Solution: Let $\gcd(a + b, a^2 - ab + b^2) = d$

$$\Rightarrow \gcd(a^2 - ab + b^2, a + b) = d, \text{ by commutative property}$$

$$\Rightarrow \gcd(a^2 - ab + b^2 - a(a + b), a + b) = d, \text{ by Ex.8}$$

$$\Rightarrow \gcd(-2ab + b^2, a + b) = d$$

$$\Rightarrow \gcd(-2ab + b^2 - b(a + b), a + b) = d, \text{ by Ex.8}$$

$$\Rightarrow \gcd(-3ab, a + b) = d \quad (1)$$

$$\Rightarrow \gcd(-3ab + 3a(a + b), a + b) = d, \text{ by Ex.8}$$

$$\Rightarrow \gcd(3a^2, a + b) = d$$

$$\Rightarrow d | 3a^2$$

$$\Rightarrow \text{Either } d = 3 \text{ or } d | a^2 \quad (2)$$

Again, from (1)

$$\gcd(-3ab + 3b(a + b), a + b) = d, \text{ by Ex.8}$$

$$\Rightarrow \gcd(3b^2, a + b) = d$$

$$\Rightarrow d | 3b^2$$

$$\Rightarrow d = 3 \text{ or } d | b^2 \quad (3)$$

From (2) and (3), we have

$$d = 3 \text{ or } (d | a^2 \text{ and } d | b^2) \quad (4)$$

$$\text{But } \gcd(a, b) = 1 \Rightarrow \gcd(a^2, b^2) = 1 \quad (5)$$

$\therefore$ In virtue of (4) and (5),

$$d = 3 \text{ or } d = 1$$

$$\therefore \gcd(a + b, a^2 - ab + b^2) = 1 \text{ or } 3$$

Ex. 10. If $a > 1$, show that

$$(a^m - 1, a^n - 1) = a^{(m,n)} - 1 \text{ for some } m, n \in \mathbb{Z}^+$$

Solution: Let $(m, n) = d$

$$\therefore d|m \text{ and } d|n$$

$$\Rightarrow m = dp, n = dq \text{ for some } p, q \in \mathbb{Z} \text{ and } (p, q) = 1$$

$$\therefore a^m - 1 = a^{dp} - 1 = (a^d)^p - 1$$

$$\therefore a^m - 1 \text{ has a factor } a^d - 1 \forall p \in \mathbb{Z}$$

Similarly $a^n - 1$ has a factor $a^d - 1 \forall q \in \mathbb{Z}$

$$\therefore a^d - 1 | a^m - 1, a^d - 1 | a^n - 1 \quad (1)$$

$$\text{Again } (a-1) | a^m - 1, (a-1) | a^n - 1 \forall a > 1. \quad (2)$$

$$\text{such that } (a-1) | a^d - 1$$

$\therefore$ In virtue of (1) and (2), we get

$$(a^m - 1, a^n - 1) = a^d - 1 = a^{(m,n)} - 1.$$

Ex. 11. Using prime factorisation, find gcd and lcm of 1300, 3575.

Also verify that $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$.

Solution. We have

$$1300 = 2^2 \cdot 5^2 \cdot 11^0 \cdot 13^1$$

$$3575 = 2^0 \cdot 5^2 \cdot 11^1 \cdot 13^1$$

$$\therefore \gcd(1300, 3575) = 2^{\min(2,0)} \cdot 5^{\min(2,2)} \cdot 11^{\min(0,1)} \cdot 13^{\min(1,1)}$$

$$= 2^0 \cdot 5^2 \cdot 11^0 \cdot 13^1 = 325$$

$$\text{lcm}(1300, 3575) = 2^{\max(2,0)} \cdot 5^{\max(2,2)} \cdot 11^{\max(0,1)} \cdot 13^{\max(1,1)}$$

$$= 2^2 \cdot 5^2 \cdot 11^1 \cdot 13^1 = 14300$$

$$\therefore \gcd(1300, 3575) \cdot \text{lcm}(1300, 3575) = 325 \times 14300$$

$$= 4647500 = 1300 \times 3575$$

$$\text{Hence } \gcd(a, b) \cdot \text{lcm}(a, b) = ab.$$

Ex. 12. Find the integers x and y such that $154x + 260y = 3$

Solution. Here $\gcd(154, 260) = 2$. Since 3 is not divisible by 2, it follows that the given equation has no integral solution.

Ex. 13. Find the gcd of 252 and 595 and express it in the form $252x + 595y$

Solution. By division algorithm,

$$595 = 2 \times 252 + 91 \quad (i)$$

$$252 = 2 \times 91 + 70 \quad (ii)$$

$$91 = 1 \times 70 + 21 \quad (iii)$$

$$70 = 3 \times 21 + 7 \quad (iv)$$

$$21 = 3 \times 7 + 0 \quad (v)$$

Since the last non-zero remainder is 7, so

$$\gcd(252, 595) = 7$$

To express the gcd in the form $252x + 595y$, we have, from (iv)

$$7 = 70 - 3 \times 21$$

$$= 70 - 3(91 - 1 \times 70), \text{ by (iii)}$$

$$= 4 \cdot 70 - 3 \cdot 91$$

$$= 4(252 - 2 \times 91) - 3 \cdot 91, \text{ by (ii)}$$

$$= 4 \cdot 252 - 11 \cdot 91$$

$$= 4 \cdot 252 - 11(595 - 2 \times 252), \text{ by (i)}$$

$$= 26 \cdot 252 - 11 \cdot 595$$

$$\therefore 7 = 252x + 595y \text{ where } x = 26, y = -11$$

Ex. 14. Find all solutions in positive integers of $5x + 3y = 52$.

Solution. Since $\gcd(5, 3) = 1$ and 52 is divisible by 1, so the equation has integral solutions.

$$\text{Now } 5 = 1 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$\therefore 1 = 3 - 1 \cdot 2 = 3 - 1(5 - 1 \cdot 3)$$

$$= -1 \cdot 5 + 2 \cdot 3$$

$$\text{Hence } 52 = 5(-52) + 3 \cdot 104$$

Thus one integral solution of the equation is

$$x_0 = -52, y_0 = 104.$$

Hence all integral solutions of the equation are given by

$$x = -52 + 3k, y = 104 - 5k \text{ for all integers } k$$

As the solution will be positive, we must have

$$-52 + 3k > 0 \text{ and } 104 - 5k > 0$$

$$\therefore \frac{52}{3} < k < \frac{104}{5}$$

$$\therefore k = 18, 19, 20.$$

Thus the given equation has exactly three positive solutions and these are

$$x = -52 + 3 \cdot 18 = 2, y = 104 - 5 \cdot 18 = 14$$

$$x = -52 + 3 \cdot 19 = 5, y = 104 - 5 \cdot 19 = 9$$

$$x = -52 + 3 \cdot 20 = 8, y = 104 - 5 \cdot 20 = 4.$$

Ex. 15. If a is a prime integer such that $a = n^2 - 4$ for some integer n , then show that $a = 5$.

Solution. We have $a = n^2 - 4 = (n - 2)(n + 2)$ for some integer n .

Since a is prime, we must have, either $n - 2 = 1$ or, $n + 2 = 1$

But $n + 2 \neq 1$ for some integer n . Hence

$$n - 2 = 1$$

$$\text{i.e., } n = 3.$$

$$\therefore a = n^2 - 4 = 3^2 - 4 = 5.$$